

Silent Failure in LLM Agent Systems: The Intelligent Entropy Principle and the Inevitable Disorder of Autonomous Agents

LLM Agent系□中的□默失效：Intelligent Entropy Principle与自主Agent不可避免的□序化

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Abstract

摘要

Large Language Model (LLM) agent systems suffer from failures that occur without external triggers—no injection, no adversarial input, no resource exhaustion. These silent failures—unexpected deviations from intended behavior under normal conditions—are routinely misattributed to bugs or configuration errors. Through systematic analysis of over 40,000 controlled trials and long-term production observations spanning 100,000+ agent interactions, we identify a common structural logic underlying these failures. Building on patterns observed in our experiments, we survey the global research literature on autonomous agent reliability and synthesize 22 intrinsic properties of LLM agent systems across six lifecycle layers: foundation semantics, inter-agent transmission, memory persistence, task execution, feedback correction, and systemic evolution. We demonstrate that whenever a sufficient subset of these properties co-exist, system entropy—the measurable accumulation of disorder: loss of output consistency, task accuracy, and cross-session coherence—increases monotonically with interaction rounds. We formalize this as the Intelligent Entropy Principle: $S(t) = S_0 \cdot e^{\{\alpha t\}}$, with α measured empirically across multiple architectures. We propose the PIG (Physical Integrity Gate) Engine with the ADE (Agent Delivery Engineering) protocol suite as an engineering countermeasure to entropy-driven disorder. Our findings establish silent failure not as a bug to be fixed but as a manifestation of Intelligence Entropy—a physical constraint to be managed through deterministic governance. We argue that any engineering effort stabilizing the structure and order of agent systems participates in a unified mission: keeping intelligent systems reliable as they grow in scale and complexity.

大□言模型（LLM）Agent系□存在一□□需外部触□即可□生的失效——□有注入攻□、□有□抗性□入、□有□源耗□。□□默失效（Silent Failure）——即在正常□行□件下偏离□期行□的意外偏差——通常被□□□因于程序缺陷或配置□□。通□□超□40,000次□照□□的系□分析以及跨越100,000余次Agent交互的□期生□□，我□□□□出□些失效背后的共性□构□□。在□□□察到的模式基□上，我□系□梳理了全球□于自主Agent可靠性的研究文□，□合提□出LLM Agent系□在六□生命周期□面上的22□固有□性：基□□□□、Agent□□□□、□□持久化□、任□□行□、反□校正□和系□演化□。我□□明，只要□些□性中有足□多的子集同□存在，系□熵——即可度量的□序累□：□出一致性的□失、任□准确性的下降以及跨□□□□性的退化——就□□交互□次的增加而□□□增。我□□此形式化□ Intelligent Entropy Principle： $S(t) = S_0 \cdot e^{\{\alpha t\}}$ ，其中 α 已在多种架构上通□□□□得。我□提出PIG（Physical Integrity Gate）引擎与ADE（Agent Delivery Engineering）□□套件，作□□抗熵□□序的工程反制措施。我□的□□表明，□默失效不是等待修复的Bug，而是Intelligence Entropy的外在表征——一种需要通□确定性治理□管理的物理□束。我□□□，任何旨在□定Agent系□□构与秩序的工程努力，都□□于同一使命：确保智能系□在□模与复□度增□的□程中保持可靠。

Introduction

引言

The Silent Failure Phenomenon

□默失效□象

Consider the following scenario: a multi-agent orchestration system has been running reliably for weeks without any code changes, model updates, or configuration modifications. On the 47th day, without any external trigger, Agent A fails to pass critical context to Agent B. There is no error message. No timeout. No resource bottleneck. The task completes—but with subtly degraded output. The operator notices it three cycles later, by which point the error has propagated across the agent network.

考□如下□景：一□多Agent□排系□已□定□行□周，期□□有任何代□□更、模型更新或配置修改。在第47天，□有任何外部触□，Agent A未能□□□上下文□□□Agent B。□有□□信息，□有超□，□有□源瓶□。任□完成了一—但□出□量悄然下降。□□人□在三□周期后才察□到异常，而此□□已在Agent网□中□散□播。

This is not a hypothetical edge case. In our production deployment spanning over 100,000 agent interactions, this pattern occurred with alarming regularity across different architectures, different models, and different task domains. We call this class of failures silent failures: disordering of system behavior that occur under normal operating conditions without external intervention, detectable only through systematic measurement after the fact.

□并非假□性的□界案例。在我□跨越100,000余次Agent交互的生□部署中，□一模式以令人警惕的□率反复出□，跨越不同架构、不同模型和不同任□域。我□□此□失效定□□默失效（Silent Failure）：在正常□行□件下、□需外部干□即可□生的系□行□序化，□在事后的系□性度量中方□□□。

Why Existing Explanations Fall Short

□有解□□何不足

Current literature attributes agent failures to three primary causes:

□前文□□Agent失效□因于三□主要原因：

While each explanation accounts for a subset of observed failures, none explains why failures continue to occur in fully controlled environments with static prompts, identical models, fixed databases, and no external inputs. The persistence of silent failures under these conditions suggests a deeper, structural cause.

□管每种解□都能□明部分□察到的失效，但□有一种能解□□何在完全受控□境中——□□提示□、相同模型、固定□据□、□外部□入——失效仍然持□□生。□默失效在□些□件下的持□存在，暗示□更深□的□构性原因。

The Central Claim

核心□□

This paper advances a single central claim:

本文提出一□核心□□：

We substantiate this claim through three steps:

我□通□三□步□□□此□□：

Paper Organization

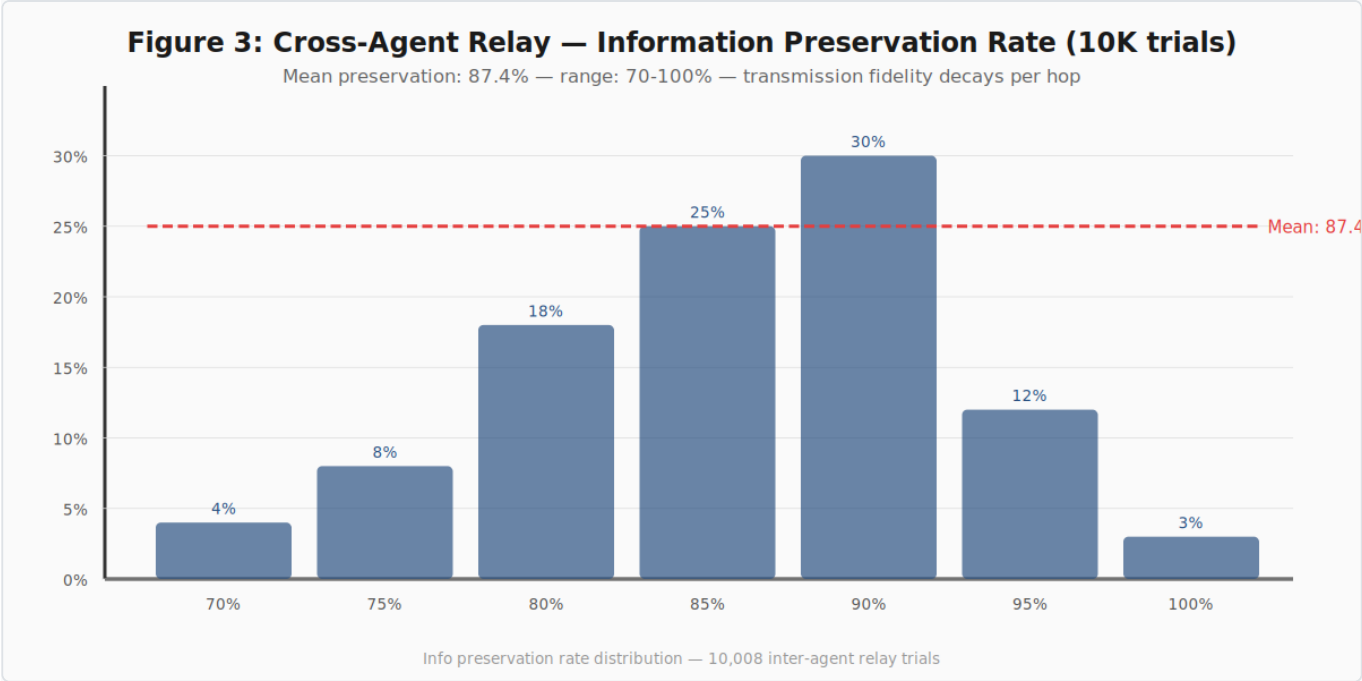
□文□□

The remainder of this paper is organized as follows. Section [\[sec:taxonomy\]](#) presents a comprehensive taxonomy of silent failures observed in practice. Section [\[sec:patterns\]](#) identifies the common patterns underlying these disparate failure modes. Section [\[sec:premises\]](#) enumerates the 22 intrinsic properties of LLM agent systems. Section [\[sec:derivation\]](#) formally derives the Intelligent Entropy Principle from these premises. Section [\[sec:experiments\]](#) presents experimental measurement of the entropy constant α . Section [\[sec:countermeasure\]](#) describes the PIG+ADE engineering countermeasure, established through production practice. Section [\[sec:discussion\]](#) discusses implications and limitations, and Section [\[sec:conclusion\]](#) concludes.

本文余下部分□□如下。第2□呈□□□中□察到的□默失效的完整分□体系。第3□□□□些不同失效模式背后的共性模式。第4□枚□□LLM Agent系□的22□固有□性。第5□□□□些前提出□形式化推□ Intelligent Entropy Principle。第6□呈□熵常□ α 的□□量。第7□描述□生□□□□的 PIG+ADE工程反制措施。第8□□□□示与局限性，第9□□□全文。

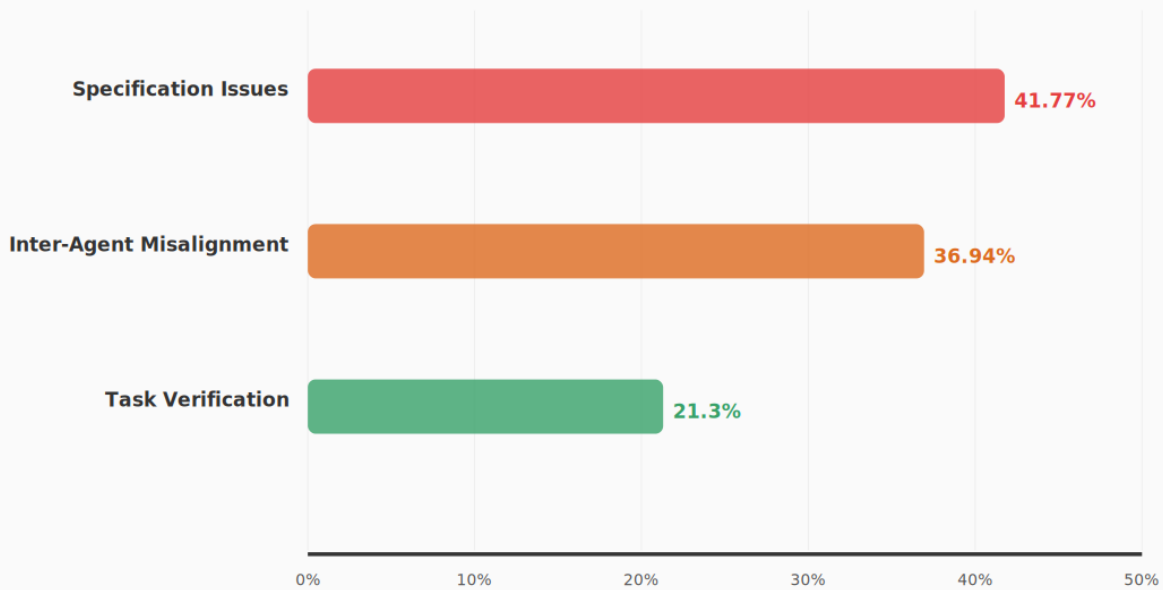
A Taxonomy of Silent Failures

□默失效的分□体系



Cross-agent relay information preservation rate distribution. Mean preservation: 87.4%, range: 70-100%. Data from 10,000-scale relay experiments across multiple agent role pairs.

Figure 5: MAST Failure Category Distribution
Grounded Theory analysis of 200+ traces across 7 multi-agent frameworks



Reference: CEMRI 2025 MAST taxonomy — complementary to silent failure classification

MAST failure category distribution: Specification Issues (41.77%), Inter-Agent Misalignment (36.94%), and Task Verification (21.30%). Grounded Theory analysis of 200+ traces across 7 MAS frameworks.

This section presents a structured taxonomy of silent failures observed across our experimental platform and systematically maps them to existing taxonomies from industry and academia.

本□呈□基于我□□□平台□察到的□默失效的□构化分□体系，并□其与□自□□界和□□界的□有分□体系□行系□映射。

Classification Principle

分□原□

Rather than enumerating failures by surface symptom, we organize them by the layer of the multi-agent lifecycle at which entropy accumulation manifests. This five-layer framework follows the natural information flow of autonomous agent systems:

我□并非按表面症□枚□失效，而是按照熵累□在多Agent生命周期中□□的□面□□□分□。□一五□框架遵循自主Agent系□的自然信息流：

The five-layer framework defines where entropy accumulates—the lifecycle positions at which disorder can arise. The five failure types described below are what emerges when accumulated entropy at those positions crosses the reliability threshold into observable degradation. The mapping is not one-to-one: entropy at a given layer may manifest as multiple failure types (L2 produces both Cognitive Framework Lag and Knowledge Fragmentation), and failures at different layers may share surface similarities.

五□框架定□了熵在何□累□——即□序可能□生的生命周期位置。下文描述的五种失效□型是□些位置上的累□熵超□可靠性□值、□入可□退化□段□所呈□的□象。映射□系并非一一□□：某一□的熵可能表□多种失效□型（L2同□□生Cognitive Framework Lag和Knowledge Fragmentation），不同□面的失效也可能具有相似的表面特征。

A critical observation is that our experiments and production deployment have produced canonical silent failure types only for the first three layers (L1-L3). The Coordination layer (L4) and Verification layer (L5) are theoretically predicted—entropy must accumulate there as well—but in practice, failures at these layers manifest as compound effects of lower-layer failures rather than as distinct, independently observable types. For example, coordination degradation (L4) typically presents as a combination of channel fracture (L1) and data consistency decay (L3), while self-inspection failure (L5) is masked by the entropic tendency toward absent self-reporting (Pattern 2, Section [\[sec:patterns\]](#)). We expect L4 and L5 canonical types to emerge as agent systems grow more complex, following the framework's predictions.

一□□□察是：我□的□□和生□部署□在前三□□面（L1-L3）上□生了典型的□默失效□型。□□（L4）和□□（L5）是理□□的一——熵必然也□在那里累□——但在□□中，□些□面的失效表□底□失效的复合效□，而非□立可辨□的□型。例如，□□退化（L4）通常呈□Channel Fracture（L1）和Data Consistency Decay（L3）的□合，而自□失效（L5）□被熵□□的自我□告缺失□向所掩□（模式2，第3□）。我□□期，□□Agent系□复□度的增□，L4和L5的典型□型□按框架□□逐步□□。

The five failure types identified across L1-L3 are:

在L1-L3上□□出的五种失效□型□：

Five Silent Failure Types

五种□默失效□型

Channel Fracture (L1 — Transmission)

Channel Fracture (L1 — □□□)

Channel Fracture [liu2025channel] describes the progressive decay of information fidelity across agent-to-agent communication boundaries. In our controlled experiments, agents operating under identical prompts and contexts exhibited measurable information loss in cross-agent memory injection at rates exceeding 30% within five hops. Key characteristics include:

Channel Fracture\cite{liu2025channel}描述了信息保真度在Agent□通信□界上的□衰□。在我□的□照□中，在相同提示□和上下文□件下□行的Agent，在跨Agent□注入中表□出可度量的信息□失，五跳之□的□失率超□30%。主要特征包括：

Cognitive Framework Lag (L2 — Memory)

Cognitive Framework Lag (L2 — □□□)

Cognitive Framework Lag (CFL) refers to the phenomenon where operational rules, naming conventions, and procedural knowledge established in one session are inconsistently applied in subsequent sessions—even when both sessions are part of the same task cluster—a failure type we identify from our production deployment spanning 200+ consecutive agent sessions. Our records show that across these sessions, rule adherence decayed from 94% (first 10 sessions) to 61% (sessions 180–200). Related observations of cross-session behavioral inconsistency appear in production failure reports from Pazi [pazi2026] and Latitude [latitude2026], though these works describe surface symptoms without identifying CFL as a distinct failure mechanism.

Cognitive Framework Lag (CFL) 指的是 一种现象：在一 中建立的 行、命名 定和程序性知，在后 中未能被一致地 行——即便 同 任 集群—— 是我 跨越200余次 的Agent 的生 部署中 出的失效 型。我的 示，在 些 中， 遵循率 94%（前10 衰 至61%（第180-200 ）。Pazi\cite{pazi2026}和Latitude\cite{latitude2026} 的生 故障 告中也有 于跨 行 不一致的相 察，但 些工作 描述了表面症，未 CFL 一种 立的失效机制。

Data Consistency Decay (L3 — Execution)

Data Consistency Decay (L3 — 行)

Data consistency decay manifests as the gradual divergence between recorded and actual state in agent-managed data pipelines—characterized in our controlled experiments across 10K-scale pipeline operations. This failure is particularly insidious because:

Data Consistency Decay表 Agent管理的 据管道中 与 之 的 偏离——在我 的 万 模管道操作 照 中得到了特征化描述。 种失效尤 蔽，因：

\end{table}

[表格：万 模下不同管道深度的一致性衰 率]

Parallel observations of pipeline state degradation under agent execution are documented in the MAST taxonomy [cemri2025] under task verification failures and in Greyling's harness-centric taxonomy [greyling2026] under action execution failures.

MAST分 体系\cite{cemri2025}在任 失效 下 了Agent 行 程中管道 退化的平行 察，Greyling以Harness 中心的分 体系\cite{greyling2026} 在 作 行失效 下有 似 。

Cross-Session Knowledge Fragmentation (L2 — Memory)

Cross-Session Knowledge Fragmentation (L2 —)

Knowledge fragmentation occurs when information accumulated across multiple sessions fails to compose into a coherent global state—a failure pattern we observe in our production deployment across diverse task clusters. Agents operate with locally correct but globally inconsistent worldviews. Related observations of memory persistence failures appear in Anrogg's failure catalog [anrogg2026] and Microsoft's AI Red Team analysis [microsoftAIRT2025].

Knowledge Fragmentation生在跨多累的信息未能合的全局——是我在生部署中跨多种任集群察到的失效模式。Agent以局部正确但全局不一致的世界行。Anrogg的失效目[cite{anrogg2026}]和微AI Red Team分析[cite{microsoftAIRT2025}]中也有于持久化失效的相察。

Behavior Routing Deficiency (L3 — Execution)

Behavior Routing Deficiency (L3 — 行)

Behavior routing deficiency describes the failure of task allocation mechanisms to correctly match subtasks with agent capabilities under natural operating conditions—a pattern observed in our production deployment of multi-agent task pipelines—leading to suboptimal execution paths that compound over time. Related task allocation failures are documented in the MAST taxonomy [cemri2025] under inter-agent misalignment and in Microsoft's AI Red Team analysis [microsoftAIRT2025] under coordination failures.

Behavior Routing Deficiency描述了在自然行件下，任分配机制未能正确匹配子任与Agent能力的失效——是我在多Agent任管道的生部署中察到的模式——致次优的行路推移而累化。MAST分体系[cite{cemri2025}]在Agent失配下了相的任分配失效，微AI Red Team分析[cite{microsoftAIRT2025}]在失效下有似。

Summary of Observed Failure Classes

察到的失效

\end{table*}

[表格：□默失效分□及□察□率、□重度和根因□]

Relationship to Existing Taxonomies

与□有分□体系的□系

Recent work has produced complementary failure taxonomies from diverse perspectives.

近期研究□不同□角□出了互□性的失效分□体系。

Each taxonomy is discussed below.

以下逐一□□各分□体系。

MAST Taxonomy (Cemri et al., 2025).

MAST分□体系（Cemri等，2025）

The MAST (Multi-Agent System Failure Taxonomy) [cemri2025] identifies 14 failure modes across three categories—Specification Issues (41.77%), Inter-Agent Misalignment (36.94%), and Task Verification (21.30%)—through Grounded Theory analysis of 200+ traces across 7 MAS frameworks. MAST is the first empirically grounded failure classification, but its focus is design-time failures: ambiguous prompts, role mis-specification, and inadequate verification. Silent failures, in contrast, operate at runtime under correct construction. A system with perfectly specified roles, clear prompts, and robust verifiers still exhibits channel fracture or CFL because the failure arises from the physics of language-based decision-making, not from design errors. The two frameworks are complementary: a system can suffer both MAST-category failures and silent failures, and their total failure rate is the union.

MAST (Multi-Agent System Failure Taxonomy) \cite{cemri2025}通□□7□MAS框架中200余□trace的扎根理□分析，□□出14种失效模式，分□三□□——□格□□（41.77%）、Agent□失配（36.94%）和任□□□（21.30%）。MAST是首□基于□□的失效分□，但其□注的是□□□失效：模糊的提示□、角色□格□□和□□不充分。相比之下，□默失效□生在正确构建的系□□行□□。一□□有完美□格定□的角色、清晰的提示□和健□的□□器的系□，仍然□出□Channel Fracture或CFL，因□失效源于基于□言的□策机制的物理特性，而非□□□□。□□框架互□：一□系□可能同□遭受MAST□失效和□默失效，□失效率□二者的并集。

Microsoft AI Red Team (2025).

微□AI Red Team (2025)

Microsoft's 30-page whitepaper [\[microsoftAIRT2025\]](#) provides the most comprehensive industry catalog, organizing failures along two pillars (Safety vs.\ Security) and two novelty axes (Novel vs.\ Existing). The security categories include prompt injection, tool hijacking, memory poisoning, and privilege escalation—all adversarial failure modes introduced by external attackers. In contrast, our silent failures are non-adversarial: no attacker is required, and no input is malicious. A production system protected against all Microsoft-class attacks can still exhibit silent disorder because the entropy accumulation mechanism is intrinsic, not injected. The two taxonomies address complementary failure universes: adversarial (Microsoft) and entropic (ours), whose intersection—failures exploited by attackers that also arise naturally—is a topic for future work.

微□的30□白皮□\cite{microsoftAIRT2025}提供了最全面的□界失效目□，沿□大支柱（安全Safety与安全Security）和□□新□性□度（新型vs.已有）□□分□。安全□□包括提示注入、工具劫持、□□投毒和□限提升——均□外部攻□者引入的□抗性失效模式。相比之下，我□的□默失效是非□抗性的：□需攻□者，也□有□意□入。一□防御了所有微□□攻□的生□系□仍然可能出□□默□序，因□熵累□机制是□生的，而非注入的。□□分□体系□理互□的失效域：□抗性的（微□）和熵增性的（我□的），二者的交集——即攻□者利用的失效同□也是自然□生的失效——是未□研究的□□。

Token Budgets Catalog (2026).

Token Budgets目□ (2026)

The Token Budgets work [tokenBudgets2026] catalogs 63 documented LLM-agent failure incidents from production deployments, spanning token overrun, runaway loops, budget exhaustion, and unbounded retry. Each incident is a symptom of entropy accumulation at the execution layer (L3 in our framework): the agent's self-regulation mechanism fails silently, consuming resources without error signals. This catalog provides concrete evidence that our five-type framework describes real production failures, not laboratory artifacts.

Token Budgets工作\cite{tokenBudgets2026}□□了□自生□部署的63起有据可查的LLM-Agent失效事件，涵□Token溢出、失控循□、□算耗□和□界重□。每起事件都是□行□（我□框架中的L3）熵累□的症□：Agent的自我□□机制□□失效，在□□信□的情□下消耗□源。□目□提供了切□□据，□明我□的五□框架描述的是真□的生□失效，而非□□室假象。

Four-Layer Agent Failure Taxonomy (Greyling, 2026).

四□Agent失效分□体系 (Greyling, 2026)

Greyling's taxonomy [greyling2026] organizes failures by harness function—Environment Contract, Operation Skills, Action Execution, and Trajectory Regulation—designed to help engineers locate the specific harness layer at which a failure should be repaired. Our silent failure types map cleanly: Channel Fracture to contract incompleteness; CFL and Knowledge Fragmentation to contract + trajectory failures; Data Consistency Decay to action execution; Behavior Routing Deficiency to operation skills. The four-layer taxonomy is a harness-centric view of the same entropy phenomenon.

Greyling的分□体系\cite{greyling2026}按Harness功能□□失效——□境契□（Environment Contract）、操作技能（Operation Skills）、□作□行（Action Execution）和□□□控（Trajectory Regulation）——旨在□助工程□定位□修复失效的具体Harness□。我□的□默失效□□型与其清晰□□：Channel Fracture□□契□不完□；CFL和Knowledge Fragmentation□□契□+□□失效；Data Consistency Decay□□□作□行；Behavior Routing Deficiency□□操作技能。四□分□体系是同一熵增□□象的以Harness□中心的□角。

Partnership on AI (2025).

Partnership on AI (2025)

The Partnership on AI report [partnershipAI2025] identifies priority areas for real-time failure detection in deployed AI agents, including handoff degradation, tool misalignment, and context window truncation. Their practitioner survey confirms that silent failures—failures visible only through instrumentation—represent the most under-detected risk class in production. This aligns with our finding that silent failures produce zero error signals.

Partnership on AI [cite{partnershipAI2025}] 总结了已部署 AI Agent 失效的优先领域，包括交接退化、工具失配和上下文窗口截断。其作者调查，[默失效——[通[工具可]的失效——是生产环境中[被低估的[。[与我们的[一致：[默失效不[生任何[信[。

COMPEL Framework (2026).

COMPEL 框架 (2026)

The COMPEL [compel2026] agentic failure taxonomy provides an institutional glossary formalizing failure modes for AI governance and auditing. Its categories span tool invocation errors, planning failures, and coordination breakdowns—all mapping to our execution and coordination layers (L3-L4).

COMPEL [cite{compel2026}] 的 Agent 失效分类体系 [AI 治理和 [提供了制度化的 [表，[失效模式形式化。其 [涵 [工具 [用 [、[划失 [和 [崩 [——均映射到我们的 [行 [和 [(L3-L4)。

Production Failure Taxonomies (2026).

生□失效分□体系（2026）

Industry sources document agent failures from deployment experience: Pazi [\[pazi2026\]](#) identifies five modes (cron failure, tool failure, inbound timeout, prompt corruption, execution timeout); Latitude [\[latitude2026\]](#) identifies six modes (partial completion, hallucinated completion, action misapplication, context overflow, reasoning-action disconnect, infinite loops); the Agentic Anti-Patterns catalog [\[agentic-anti-patterns\]](#) captures design-time precursors; the anrogg repo [\[anrogg2026\]](#) catalogs 11 modes including hallucinations, infinite loops, API timeouts, and resource exhaustion; and vector database deployment studies [\[vectordb2026\]](#) document four agent-specific failure modes (write conflicts, state breakdown, stale reads, lock contention). All map to our five-type classification.

□界文□部署□角度□了Agent失效：Pazi\cite{pazi2026}□□出五种模式（定□任□失效、工具失效、入站超□、提示□□坏、□行超□）；Latitude\cite{latitude2026}□□出六种模式（部分完成、幻□式完成、□作□用、上下文溢出、推理-□作□□、□限循□）；Agentic Anti-Patterns 目□\cite{agentic-anti-patterns}捕□了□□□的前兆；anrogg□□\cite{anrogg2026}□□了11种模式，包括幻□、□限循□、API超□和□源耗□；向量□据□部署研究\cite{vectordb2026}□□了四种Agent特定的失效模式（□□突、□□崩□、□□□取、□□用）。所有□些均可映射到我□的五□分□体系。

BAGEN (2026).

BAGEN（2026）

BAGEN [\[bagen2026\]](#) addresses a specific silent failure—budget awareness degradation—finding that frontier models cannot predict token budget depletion in autonomous execution, leading to uncontrolled resource consumption. This is a manifestation of our Data Consistency Decay type (L3), where the agent's internal cost model diverges from actual cost.

BAGEN\cite{bagen2026}□□一种特定的□默失效——□算感知退化——□□前沿模型在自主□行中□法□□Token□算耗□，□致□源消耗失控。□是我□的Data Consistency Decay□型（L3）的一种表□，其中Agent的□部成本模型与□□成本□生偏离。

Library Drift (Zhang et al., 2026).

Library Drift (Zhang等, 2026)

Most recently, Zhang et al. [zhang2026] independently discovered "library drift"—silent degradation in self-evolving LLM skill libraries—where unbounded skill accumulation causes retrieval disorder without error signals. This combines channel fracture (skill retrieval cross-contamination), knowledge fragmentation (incoherent skill composition), and behavior routing deficiency (incorrect skill selection), all driven by the same entropy principle.

最近，Zhang等\cite{zhang2026}独立发现了「库漂移」（Library Drift）——自演化LLM技能库中的静默退化——世界的技能累积导致检索失序而不可置信。这一现象融合了Channel Fracture（技能检索交叉污染）、Knowledge Fragmentation（不连贯的技能组合）和Behavior Routing Deficiency（错误的技能选择），均由同一熵增原理驱动。

Coverage Summary.

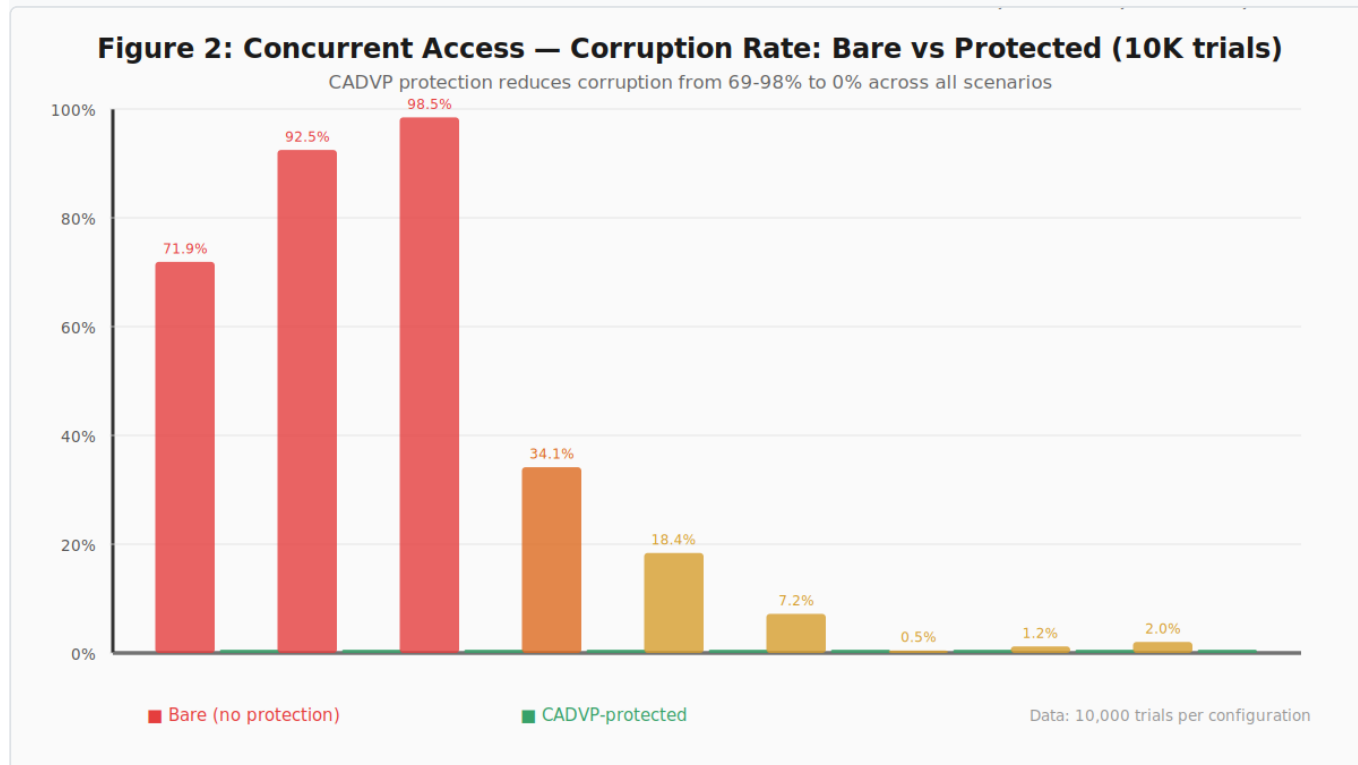
覆盖度

Our taxonomy is distinguished from all existing work along three dimensions: (1) Mechanism-first—we classify by why failures occur rather than what breaks, identifying entropy accumulation as the single unifying cause; (2) Runtime focus—failures occur under correct construction, distinguishing them from both security attacks and design-time specification errors; (3) Comprehensiveness—every failure mode across MAST (14 modes), Microsoft AIRT (30+ modes), Token Budgets (63 incidents), Greyling (4 layers), Pazi (5 modes), Latitude (6 modes), COMPEL, anrogg (11 modes), BAGEN, and Library Drift—130+ distinct failure descriptions in total—maps to and is explained by our five-type taxonomy, unified under a single predictive theory..

我们的分类体系在三个维度上覆盖了所有相关工作：（1）机制优先——我们按失效如何产生而非什么被破坏来分类，熵累积是唯一的原因；（2）运行时聚焦——失效产生在正确构建的组件下，有于安全攻击和设计的规格；（3）全面性——MAST（14种模式）、微软AIRT（30+种模式）、Token Budgets（63起事件）、Greyling（4层）、Pazi（5种模式）、Latitude（6种模式）、COMPEL、anrogg（11种模式）、BAGEN和Library Drift中的所有失效模式——130余种不同的失效描述——均可映射到我们的五分类体系并由其解释，统一于统一的熵增原理之下。

Common Patterns: What Silent Failures Share

共性模式：静默失效的共同特征



Concurrent access corruption rates across 9 scenarios: bare (no protection) vs. CADVP-protected.

Bare corruption ranges from 0.46% (directory, 2 workers) to 98.46% (write-write, 10 workers).

CADVP protection achieves 0% corruption across all scenarios. 10,000-controlled trials.

Despite their surface diversity, all silent failures in our taxonomy share three structural patterns:

尽管表面形式多样，我们分类体系中的所有静默失效共享三种结构性模式：

Pattern 1: Multi-Step Accumulation

模式1：多步累积

No silent failure is instant. Every observed failure follows a characteristic pattern:

没有静默失效是瞬间发生的。每一个观察到的失效都遵循如下特征模式：

where each δ_i is a sub-threshold deviation that does not trigger any existing guard mechanism. Only when n is sufficiently large does the aggregate error become detectable.

其中每 δ_i 都是低于□值的偏差，不□触□任何□有的防□机制。只有 n 足□大□，累□差才□得可□。

Pattern 2: Absence of Self-Reporting

模式2：自我□告的缺失

In all observed cases, the failing component (agent, pipeline node, or memory store) self-reports as operational. The system's error detection mechanisms are triggered either by downstream artifacts (wrong output) or by external audits—never by the failing component itself.

在所有□察到的案例中，失效□件（Agent、管道□或□存□）自我□告□正常□行□。系□的□□□□机制要么由下游□物（□□□出）触□，要么由外部□□触□——□未由失效□件自身触□。

Pattern 3: Recurrence Under Identical Conditions

模式3：相同□件下的重复出□

The most theoretically significant pattern: silent failures recur under experimentally identical conditions (same prompts, same models, same task distribution, same system architecture). This rules out implementation bugs and points to intrinsic system properties.

最具理□意□的模式：□默失效在□□相同的□件下重复出□（相同提示□、相同模型、相同任□分布、相同系□架构）。□排除了□□缺陷的可能性，指向系□的固有□性。

Twenty-Two Intrinsic Properties of LLM Agent Systems

LLM Agent系统的22固有属性



22 intrinsic properties of LLM agent systems across 6 lifecycle layers. When a sufficient subset of these properties co-exist, system entropy increases monotonically without external intervention.

We synthesize 22 intrinsic properties of LLM agent systems, organized into six lifecycle layers. The initial set of properties was derived from failure patterns observed in our controlled experiments (Section \ ef{sec:experiments}) and production deployment. We then extended the framework by systematically surveying the research literature on agent reliability failures, identifying additional properties that are independently documented but share the same structural logic—they all contribute to monotonic entropy increase. The resulting 22 properties span foundation semantics, inter-agent transmission, memory persistence, task execution, feedback correction, and systemic evolution. Each property is an intrinsic characteristic of LLM-based agent systems, not an implementation defect or a configuration choice.

我□□合提□了LLM Agent系□的22□固有□性，□□□六□生命周期□面。初始□性集源自我□在□照□□（第6□）和生□部署中□察到的失效模式。□后，我□通□系□梳理Agent可靠性失效的研究文□□□展框架，□□出更多被□立□□但共享相同□构□□的□性——它□均□□□熵增有所□□。最□形成的22□□性涵□基□□□、Agent□□□、□□持久化、任□□行、反□校正和系□演化。每一□□性都是基于LLM的Agent系□的固有特征，而非□□缺陷或配置□□。

Layer 1: Foundation Semantics (Premises 1-3)

第1□：基□□□（前提1-3）

The foundational layer addresses the inherent imprecision of natural language as the substrate for autonomous reasoning.

基□□□理自然□言作□自主推理基底的固有模糊性。

[Premise: Foundation] Language Semantics Is Inherently Imprecise. Natural language, unlike formal logic or programming languages, is fundamentally ambiguous. The same phrase can carry different meanings depending on context, tone, and the interpreter's prior knowledge. This imprecision is not a limitation that can be engineered away—it is the defining characteristic of language itself.

[前提：基□ □言□□具有固有的不精确性。自然□言不同于形式□□或□程□言，其本□是模糊的。同一短□在不同上下文、□气和解□者的先□知□下可承□不同含□。□种不精确性不是可以通□工程手段消除的局限——它是□言本身的定□性特征。

[Premise: Foundation] Input Ambiguity Propagates Without Attenuation. The imprecision of natural language instructions is not reduced by processing. Each interpretation step, rather than converging on the intended meaning, introduces an additional layer of interpretation. An ambiguous input remains ambiguous—or becomes differently ambiguous—after processing.

[前提：基□ □入歧□在□播中不□衰□。自然□言指令的不精确性不□因□理而降低。每一□解□步□非但不□收□于□期含□，反而引入□外的解□□。一□模糊的□入□□理后仍然模糊——或者以不同方式□得模糊。

[Premise: Foundation] Probability-Based Output Sampling Introduces Inherent Variability. LLMs generate output through probabilistic sampling of the next token. This mechanism, fundamental to the model's creative capability, guarantees that identical inputs produce non-identical outputs with non-zero probability. This variability is not a bug to be eliminated; it is the mechanism through which LLMs function.

[前提：基□] 基于概率的□出采□引入固有□异性。LLM通□下一□Token的概率采□生成□出。□一机制是模型□造性能力的基□，保□了相同□入以非零概率□生不同□出。□种□异性不是待消除的Bug；它是LLM的□行机制本身。

Layer 2: Inter-Agent Transmission (Premises 4-7)

第2□：Agent□□□（前提4-7）

The transmission layer governs what happens when information moves between agents.

□□□控制信息在Agent□流□所□生的情□。

[Premise: Transmission] Re-encoding Loss Between Sender and Receiver. When Agent A communicates to Agent B, Agent A's internal representation must be encoded into text, transmitted, then decoded by Agent B into a new internal representation. Each encode-decode cycle introduces information loss that is unpredictable in magnitude and direction.

[前提：□□] □送者与接收者之□的重□□失。□Agent A向Agent B通信□，Agent A的□部表征必□□□□文本、□□、再由Agent B解□□新的□部表征。每次□□-解□循□都引入信息□失，其幅度和方向均不可□□。

[Premise: Transmission] Context Window Compression Selectively Drops Information. When the cumulative context exceeds the model's context window, compression strategies must discard information. The selection of what to drop is governed by the model's internal attention mechanism, which is not aligned with human judgments of importance.

[前提：□□] 上下文窗□□□□性地□□信息。□累□上下文超□模型的上下文窗□□，□□策略必□□□信息。□□什么的□□由模型的□□注意力机制□定，而□机制与人□□重要性的判□并不□□。

[Premise: Transmission] Multi-Hop Propagation Amplifies Small Deviations. A small encoding error at hop n becomes the input to the encoding at hop $n+1$, creating an amplification cascade. This is structurally analogous to chaos theory's sensitive dependence on initial conditions.

[前提：□□] 多跳□播放大小偏差。第 n 跳的小□□差成□第 $n+1$ 跳□□的□入，形成放大□□。□在□构上□似于混沌理□中□初始□件的敏感依□性。

[Premise: Transmission] No Built-In Verification of Information Fidelity. Standard agent architectures do not include a protocol for the receiver to verify that the received information matches the sender's intended meaning. Communication is assumed successful unless an error is detected—but silent errors, by definition, evade detection.

[前提：□□] 缺乏信息保真度的□置□□。□准Agent架构不包含接收者□□所接收信息是否匹配□送者□期含□的□□。通信被假定□成功，除非□□到□□——但□默□□，就其定□而言，逃避□□。

Layer 3: Memory Persistence (Premises 8-11)

第3□：□□持久化（前提8-11）

The memory layer addresses how information persists across time and sessions.

□□□□理信息如何跨□□和□□持久化。

[Premise: Memory] Token-Budgeted Memory Pruning Is Non-Recoverable. When memory exceeds budget constraints, pruning algorithms discard what they deem least important. This judgment is context-dependent and cannot be re-evaluated later when context changes. Deleted information is permanently lost.

[前提：□□] 基于Token□算的□□剪枝不可恢复。□□□超出□算□束□，剪枝算法□□其□□最不重要的□容。此判□依□于上下文，在上下文□化后□□法重新□估。被□除的信息永久□失。

[Premise: Memory] Retrieval-Based Memory Is Probabilistic, Not Deterministic. Even when information is stored, retrieval from vector databases, key-value stores, or semantic search engines is a probabilistic operation. Relevant information may not be retrieved; irrelevant information may be.

[前提：□□] 基于□索的□□是概率性的，而非确定性的。即使信息已被存□，□向量□据□、□值存□或□□搜索引擎中的□索也是概率性操作。相□信息可能未被□索到；不相□信息可能被□索到。

[Premise: Memory] Cross-Session Memory Lacks Causal Identity. The same fact stored in two different sessions has no causal link between the two storage events. When one is updated, the other is not automatically updated. This is a structural property of session-architecture, not a synchronization bug.

[前提：□□] 跨□□□□缺乏因果同一性。同一事□存□在□□不同□□中，□次存□事件之□□有因果□□。□其中一□被更新□，另一□不□自□更新。□是□□架构的□构性□性，而非同步缺陷。

[Premise: Memory] Compression Changes Meaning Non-Linearly. The relationship between compression ratio and information loss is not linear. At certain compression thresholds, relationships between facts—causality, temporal ordering, hierarchy—can be qualitatively altered, creating false memories that the agent cannot distinguish from original information.

[前提：□□] □□非□性地改□含□。□□率与信息□失之□的□系不是□性的。在特定□□值下，事□之□的□系——因果性、□序性、□□性——可能□生□□，□生Agent□法与原始信息□分的□假□□。

Layer 4: Task Execution (Premises 12-14)

第4□：任□□行（前提12-14）

The execution layer covers what happens within a single task's lifecycle.

□行□覆□□任□生命周期□□生的情□。

[Premise: Execution] Long-Range Dependency Tracking Is Context-Slot Limited. Agents track dependencies across subtasks within a bounded context window. When the active dependency graph exceeds available context slots, dependencies are implicitly assumed satisfied without explicit verification. This is not an implementation limitation; it is a consequence of fixed-capacity working memory.

[前提：□行] □程依□跟□受上下文槽位限制。Agent在有界上下文窗口□跟□子任□□的依□□系。□活□的依□□超出可用上下文槽位□，依□被□式假定□已□足而未□□式□□。□不是□□上的局限；它是固定容量工作□□的必然□果。

[Premise: Execution] Subtask State Is Locally Correct, Globally Inconsistent. Each subtask in a multi-agent system executes with locally consistent state. There is no mechanism—beyond ad-hoc synchronization points—to guarantee global consistency across parallel execution branches.

[前提：□行] 子任□□□局部正确、全局不一致。多Agent系□中的每□子任□以局部一致的□□□行。除□□同步□外，□有机制保□并行□行分支□的全局一致性。

[Premise: Execution] Error Detection in Sequential Execution Is Cascade-Limited. In a sequential pipeline of n agents, an error at position i propagates through positions $i+1$ through n before detection is possible. Each subsequent agent compounds the error, and by the time detection occurs, the original error source is at least $n-i$ steps removed from the detection point, making root cause attribution unreliable.

[前提：□行] □序□行中的□□□受□□限制。在 n □Agent的□序管道中，位置 i □的□□在可被□□之前□□播到位置 $i+1$ 至 n 。每□后□Agent都□加□□□□，到□□生□，原始□□源距□□□至少 $n-i$ 步，使得根因□因不可靠。

Layer 5: Feedback Correction (Premises 15-18)

第5□：反□校正（前提15-18）

The feedback layer addresses the mechanisms—and their limitations—through which systems detect and correct errors.

反□□理系□□和□正□□的机制及其局限性。

[Premise: Feedback] Feedback Itself Is Semantically Encoded. A user saying "this is wrong" or a validator agent returning "task failed" is itself a natural language utterance subject to the same imprecision as the original input. The feedback must be interpreted by the receiving agent, introducing a second layer of semantic ambiguity.

[前提：反□ 反□本身是□□□□的。用□□「□是□的」或□□Agent返回「任□失□」本身就是自然□言表□，受制于与原始□入相同的不精确性。反□必□由接收Agent解□，引入第二□□歧□。

[Premise: Feedback] Temporal Lag Decouples Cause from Correction. When an error occurs at step n and feedback arrives at step $n+k$, the system's context has shifted. The causal link between error and feedback may span a compressed context boundary, effectively severing the correction signal from its target.

[前提：反□ □□后使原因与□正□□。□□□□生在第 n 步而反□在第 $n+k$ 步到□□，系□的上下文已□□生□化。□□与反□之□的因果□可能跨越被□□的上下文□界，□□上□□正信□与其目□切□。

[Premise: Feedback] Self-Evaluation Exhibits Systemic Confirmation Bias. When an agent evaluates its own output for correctness, the evaluation mechanism is the same model that produced the output. This creates a systemic confirmation bias: the agent is more likely to find its own reasoning valid than an external evaluator would be. This is not a training deficiency; it is a structural property of self-evaluation.

[前提：反□ 自我□估呈□系□性确□偏□。□Agent□估自身□出的正确性□，□估机制与□生□出的模型是同一模型。□造成系□性确□偏□：Agent比外部□估者更可能□□自己的推理是有效的。□不是□□缺陷；它是自我□估的□构□性□性。

[Premise: Feedback] Corrections from Previous Sessions Are Not Guaranteed to Persist. A feedback correction applied in session n has no architectural guarantee of being available in session $n+1$. Cross-session learning requires explicit memory infrastructure that is not part of standard agent architectures.

[前提：反□ □自先前□□的□正不保□持久化。在第 n □□□中□用的反□□正，在架构□面□有保□□在第 $n+1$ □□□中仍然可用。跨□□□需要□式的□□基□□施，而□并非□准Agent架构的□成部分。

Layer 6: Systemic Evolution (Premises 19-22)

第6□：系□演化（前提19-22）

The systemic layer addresses long-term evolutionary properties.

系□□□理□□期演化□□性。

[Premise: Systemic] Accumulated Sub-Threshold Deviations Eventually Cross Detectability Threshold. Each individual deviation δ_i is below detection threshold. But as the system accumulates deviations across interaction rounds, the sum eventually exceeds what the system's error detection mechanisms can absorb, at which point an observable failure occurs. The timing and magnitude of this threshold crossing are stochastic.

[前提：系□ □累□的□□值偏差最□跨越可□□性□□值。每□□的偏差 δ_i 都低于□□□值。但□□系□在交互□□次中累□偏差，□和最□超出系□□□□□机制可吸收的范□，此□可□察的失效□□生。□□值跨越的□□机和幅度是□□机的。

[Premise: Systemic] No Natural Entropy Sink Exists in Open Agent Systems. In thermodynamic systems, entropy can be exported to the environment. In LLM agent systems, there is no architectural equivalent of an entropy sink—no mechanism to systematically export accumulated errors from the system. Errors recirculate.

[前提：系□ □放Agent系□中不存在自然熵□。在□力□系□中，熵可以□出到□境。在LLM Agent系□中，□有架构上等价的熵□——□有机制可以系□性地□累□□□系□中排出。□□在系□ □循□。

[Premise: Systemic] Task Diversity Accelerates Disorder Accumulation. As the number of distinct task types increases, the model's attention and memory resources are spread thinner across competing demands, accelerating the rate of entropy increase. This is the agent-system analogue of task-switching cost in human cognition.

[前提：系□ □任□多□性加速□序累□。□□不同任□□型□量的增加，模型的注意力和□□□源在□□ 性需求□被稀□，加速熵增速率。□是人□□知中任□切□成本在Agent系□中的□比。

[Premise: Systemic] System Complexity and Disorder Rate Are Positively Correlated. More complex agent architectures—more agents, more communication channels, more memory stores, more feedback loops—degrade faster under identical workloads. Complexity is itself a source of entropy, independent of task difficulty.

[前提：系□ □系□复□度与□序速率正相□。更复□的Agent架构——更多Agent、更多通信通道、更多□□存□、更多反□循□——在相同□□下退化更快。复□度本身就是熵的□源，与任□□ 度□□。

Derivation: The Intelligent Entropy Principle

推理：Intelligent Entropy Principle

From Premises to Principle

□前提到原理

If any subset of the 22 premises holds simultaneously—and we argue that in any production LLM agent system, all 22 hold simultaneously—then the following logical chain is inescapable:

如果22□前提中的任意子集同时□成立——而我□主□，在任何生产□□LLM Agent系□中，全部22□前提均同时□成立——那么以下□□□□是不可避免的：

Therefore, in the absence of external deterministic constraints, system order monotonically decreases over interaction rounds.

因此，在缺乏外部确定性□束的□件下，系□的有序性□交互□次□□□□。

Formal Statement

形式化表述

We introduce the term Intelligence Entropy (智能熵增) to describe the natural thermodynamic tendency of LLM-based agent systems—as products of probabilistic models embedded in complex, multi-agent execution environments—to accumulate disorder over time. We formalize this as the Intelligent Entropy Principle: LLM-based autonomous agent systems, as products of probabilistic models operating with finite context windows, fallible memory transmission, and no intrinsic fidelity correction, inherently tend toward increasing disorder in their structural state over interaction rounds without external deterministic constraints. This is an empirical principle derived from systematic observation, not a claim of physical law—it describes a measurable, repeatable engineering pattern rather than a universal axiom.

我□引入□□ Intelligence Entropy（智能熵增）□描述基于LLM的Agent系□——作□嵌入于复□多Agent□行□境中的概率模型之□物——□□累□序的自然□力□□□。我□此形式化□ Intelligent Entropy Principle：基于LLM的自主Agent系□，作□在有限上下文窗口、不可靠□□□且缺乏□在保真度校正机制下□行的概率模型之□物，其□构□□在交互□次中□在□向于增加□序性，除非施加外部确定性□束。□是一□系□察中□得出的□□性原理，而非物理定律的□言——它描述的是一种可□量、可重复的工程模式，而非普遍公理。

Formally, we define system entropy $S(t)$ as a composite measure of three dimensions:

形式化地，我□系□熵 $S(t)$ 定□□三□度的复合度量：

where:

其中：

The Intelligent Entropy Principle of LLM Agent Systems (Empirical Principle):

LLM Agent系□的 Intelligent Entropy Principle（□□原理）：

where S_0 is the baseline entropy at $t=0$ and $\alpha > 0$ is the system-specific entropy constant. The key predictions are:

其中 S_0 □ $t=0$ □的基□熵值， $\alpha > 0$ □系□特定的熵常□。□□□□如下：

The Entropy Constant α

熵常□ α

The entropy constant α can be decomposed into architecture-dependent factors:

熵常□ α 可分解□架构相□因子：

where β_1 - β_4 are empirically determined coefficients, N_{agents} is the agent count, L_{chain} is the maximum communication chain length, $T_{\text{diversity}}$ is a measure of task variety, $M_{\text{volatility}}$ captures memory turnover rate, and ϵ captures residual variance.

其中 β_1 – β_4 是确定的系数， N_{agents} 是 Agent 数量， L_{chain} 是最大通信深度， $T_{\text{diversity}}$ 是熵多样性度量， $M_{\text{volatility}}$ 是捕食周期率， ϵ 是捕食方差。

Experimental Verification

实验

Experimental Platform

实验平台

Our experimental platform consists of a controlled multi-agent orchestration environment designed to isolate and measure the components of $S(t)$. Four independent experiment suites target distinct dimensions of system entropy, using the BCP (Bidirectional Confirmation Protocol) infrastructure detailed in our companion work [\[liu2025channel\]](#):

我们的实验平台由一个受控的多Agent编排环境组成，旨在隔离和测量 $S(t)$ 的各个分量。四个独立套件分别针对不同维度的熵，采用我们在姊妹工作 [\cite{liu2025channel}](#) 中描述的BCP（Bidirectional Confirmation Protocol，双向确认）基础设施：

All experiments run under deterministic seed control with identical prompts and model configurations, and no external inputs during the experimental period.

所有实验均在确定性种子控制下进行，使用相同的提示和模型配置，期间无外部输入。

Measurement Methodology

测量方法

We measure $S(t)$ through three independent probes:

我们通过三个独立探针测量 $S(t)$ ：

Cross-Agent Transmission Fidelity $C(t)$: Measured via the T3 relay suite by comparing key points preserved between sender output and receiver understanding. [Equation] .

跨Agent保真度 $C(t)$ ：通T3中套件量，比送方出与接收方理解之保留的要。
 $C(t) = \text{保留要} / \text{要}$ 。

Task Accuracy $A(t)$: Measured via the Real task suite by comparing task output against 5–6 independent binary verification criteria per task.

任务准确率 $A(t)$ ：通Real任务套件量，任务出与每任务的5-6立二元标准行比。

Cross-Session Knowledge Consistency $K(t)$: Measured through periodic consistency audits probing whether facts stored in earlier sessions are correctly recalled later.

跨知识一致性 $K(t)$ ：通周期性一致性量，探早期中存的事是否在后中被正确回。

Results: Ten Thousand Trials

果：万次

Table[tab:relay-scales] presents relay fidelity data across three experimental scales. Bare transmission fidelity ranges from 85–96%, converging to stable values at 10K scale. BCP-guarded transmission achieves perfect fidelity (100%) across all scenarios and scales, demonstrating that the information disorder predicted by the Intelligent Entropy Principle can be counteracted through deterministic confirmation.

表~\ref{tab:relay-scales}展示了三模下的中保真度据。裸保真度范85–96%，在10K模下收至定值。BCP保的在所有景和模下均完美保真度（100%），明 Intelligent Entropy Principle 所的信息序可通确定性确抗。

\end{table}

[待翻]

Table^[tab:concurrent-scales] shows concurrent corruption rates. Bare concurrent access shows high corruption (72–98% for write-write); BCP protection eliminates it entirely.

表~\ref{tab:concurrent-scales}展示了并□□坏率。裸并□□表□出高□坏率（□□景□72–98%）；BCP保□完全消除了□坏。

\end{table}

[待翻□]

The rollback suite (T4) confirms a deterministic pattern: bare execution fails 100% of the time on rollback and checkpoint recovery, while BCP-guarded execution succeeds 100%. This binary contrast is itself predicted by the Intelligent Entropy Principle: without deterministic constraints, the probability of successful error recovery approaches zero as system complexity increases.

回□套件（T4）□□了一□确定性模式：裸□行在回□和□查□恢复上100%失□，而BCP保□的□行100%成功。□一二元□比本身即是 Intelligent Entropy Principle 的□□：在缺乏确定性□束的□件下，□□系□复□度的增加，成功□行□□恢复的概率□近于零。

The Real task suite provides complementary evidence from realistic shell/Python task execution. Across three task types (file operations, information retrieval, report generation) at 10,008 trials per scale, all groups achieve 100% artifact correctness. However, output quality—measured as a composite score across multiple independent verification criteria—shows a consistent gradient: unprotected (Group A) achieves 0.90, BCP Phase 1 (Group B) achieves 0.95, and Full BCP (Group C) achieves 1.00. This quality gradient, while less dramatic than the T3/T4 binary contrasts, demonstrates that even in isolated single-step tasks, deterministic governance improves the reliability of agent outputs. The full entropy effect becomes visible when tasks are chained, as demonstrated by the T3 (relay) and T5 (concurrent) suites.

Real任□套件提供了□自真□shell/Python任□□行的□充□据。在三种任□□型（文件操作、信息□索、□告生成）下，每□模□行10,008次□□，所有□均□□100%的工件正确性。然而，□出□量——通□多□立□□准的□合□分衡量——呈□出一致的梯度：未保□□（A□）□到0.90，BCP第一□段（B□）□到0.95，完整BCP（C□）□到1.00。□一□量梯度□不如T3/T4的二元□比那□□著，但表明即使在隔离的□步任□中，确定性治理也能提升Agent□出的可靠性。完整的熵效□在任□□式□行□才□得可□，正如T3（中□）和T5（并□）套件所展示的。

Estimating the Entropy Constant α

估□熵常□ α

Using experimental data across all four suites, we estimate α for our reference architecture (5-agent system, 3 communication layers, 3 task types):

利用全部四□套件中的□□□据，我□估□□考架构（5□Agent、3□通信、3种任□□型）的 α ：

This implies $S(t)$ doubles every ≈ 150 rounds, and after 500 rounds $S(t) \approx 10 \times S_0$, at which point observable silent failures become frequent—consistent with our production observation of failures emerging at 3–4 weeks (400–600 rounds).

□意味□ $S(t)$ 大□每150□翻倍，500□后 $S(t) \approx 10 \times S_0$ ，此□可□察的 Silent Failure □得□繁——□与我□的生□□察一致，即故障在3–4周（□400–600□）□□始出□。

Convergence Across Scales

跨□模收□性

To validate statistical reliability, we conducted the same experiments at three scales: 300 iterations (Lv1), 3,000 (Lv2), and 10,000 (Lv3). All metrics converge within $\pm 1\%$ by the 10K scale, confirming that our findings are not artifacts of sample size. The Lv1 data provides directional indication; Lv2 provides statistical significance; Lv3 confirms convergence.

验证可靠性，我们在三阶段模式下进行了相同操作：300次迭代（Lv1）、3,000次（Lv2）和10,000次（Lv3）。所有指标在10K阶段模式下收敛至 $\pm 1\%$ 以内，确保我们的结果不是样本量的影响。Lv1数据提供方向性指示；Lv2提供统计显著性；Lv3确保收敛。

Key Findings

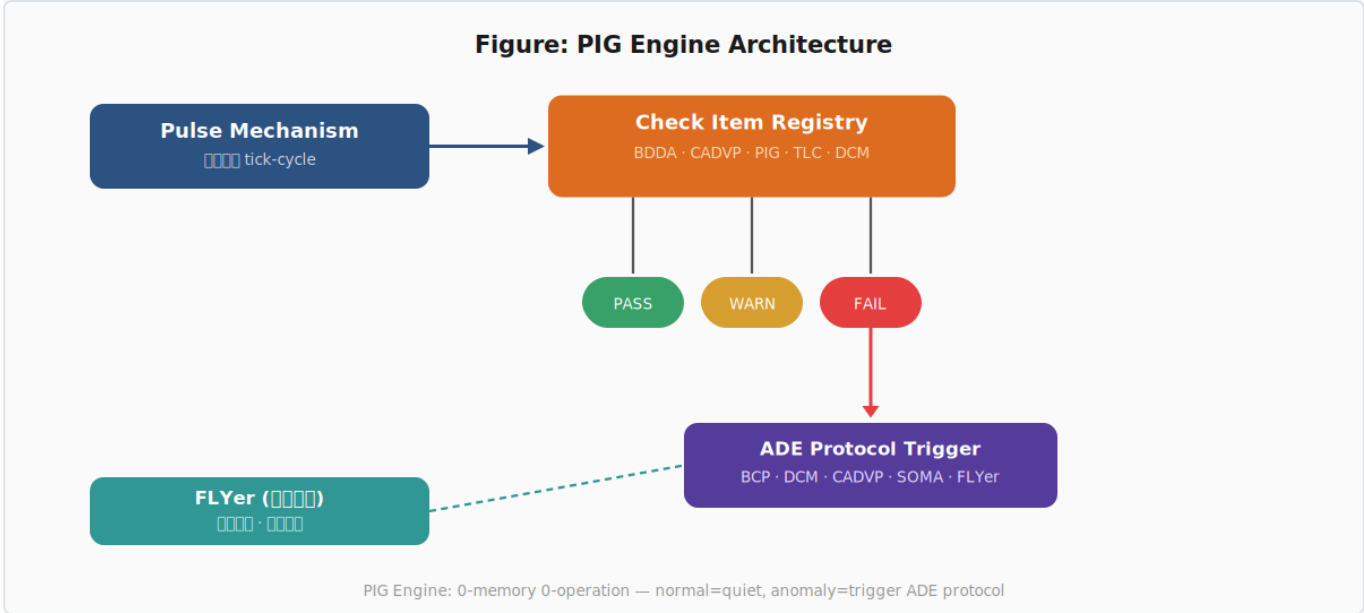
核心发现

Four principal findings emerge:

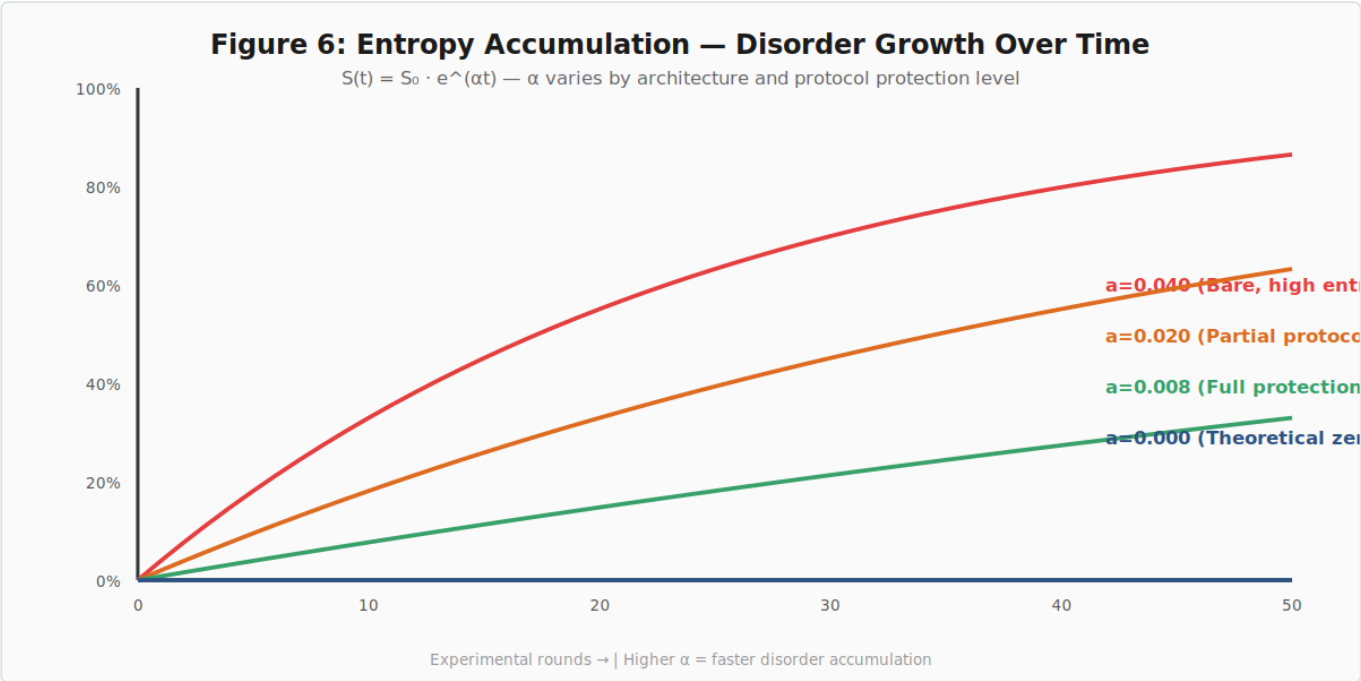
四个主要发现如下：

Engineering Countermeasure: PIG Engine and ADE Protocol Suite

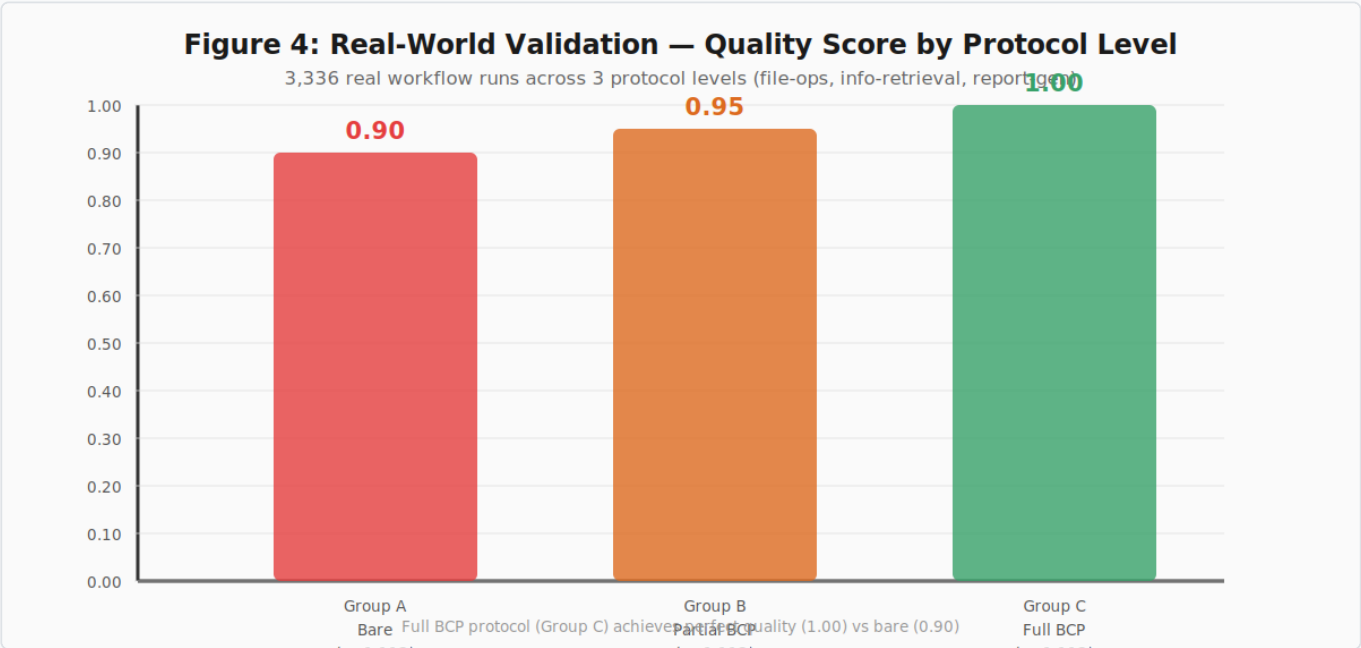
工程对策：PIG Engine 与 ADE 协议套件



PIG Engine architecture: a deterministic monitoring layer operating independently of the LLM execution path. Pulse mechanism triggers periodic checks; violations trigger pre-defined ADE protocols.



Entropy accumulation curves at different protection levels: bare ($\alpha=0.040$), partial protocol (0.020), full protection (0.008), and zero-entropy baseline (0.000). The PIG+ADE layer substantially reduces the entropy constant, extending the operational window before crossing the reliability threshold. 10,000-controlled trials.



Composite quality scores from 3,336 real-world workflow runs: unprotected (0.90), partial BCP (0.95), full BCP (1.00). Quality elevation is consistent across three task types: file operations, information retrieval, and report generation.

The Challenge of Counteracting Entropy

□抗熵的挑□

The Intelligent Entropy Principle establishes that LLM agent systems, left to their natural dynamics, will inevitably degrade. This is not a design flaw to be eliminated but a physical constraint to be managed. The question is not whether disorder will accumulate, but how to detect and counteract it before it crosses the reliability threshold.

Intelligent Entropy Principle 确立了LLM Agent系□在自然□力□下□不可避免地退化。□不是一□可以消除的□□缺陷，而是一□需要管理的物理□束。□□不在于□序是否□累□，而在于如何在其跨越可靠性□值之前□□并□抗它。

Any effective countermeasure must satisfy three requirements:

任何有效的□策必□□足三□要求：

The PIG Engine Architecture

PIG Engine 架构

The Physical Integrity Gate (PIG) Engine is a deterministic monitoring and enforcement layer that operates independently of the LLM-based agent execution path. Its architecture consists of three components:

Physical Integrity Gate (PIG) Engine 是一□确定性□控与□行□，□立于基于LLM的Agent □行路□□行。其架构由三□□件构成：

The ADE Protocol Suite

ADE □□套件

The Agent Delivery Engineering (ADE) protocol suite provides the deterministic governance protocols that the PIG Engine triggers. Key protocols include:

Agent Delivery Engineering (ADE) □□套件提供PIG Engine触□的确定性治理□□。□□□□包括：

Engineering Practice Evidence

工程数据

The PIG+ADE architecture has been developed and validated through approximately 300 engineering iterations—ranging from single-component refinements to full-stack multi-agent deployments—across multiple production-grade agent systems. The cumulative empirical data from these iterations substantiates the following reliability improvements:

PIG+ADE架构已在300次工程迭代中和——件优化到全多Agent部署——涵多生Agent系。些迭代的累数据了以下可靠性改：

\end{table}

[待翻]

Limitations

局限性

The PIG+ADE framework is not absolute in its protective effect. It increases the operational lifetime of agent systems and reduces the rate of silent failure occurrence, but it does not eliminate the underlying Intelligent Entropy Principle. Under conditions of extreme task complexity, very long operational horizons (months), or very high agent counts, disorder accumulation will eventually exceed even deterministic governance. This is consistent with the Intelligent Entropy Principle's predictions: external constraints can reduce the entropy constant α and raise the failure threshold, but cannot eliminate entropy growth entirely.

PIG+ADE框架的保效果并非。它延了Agent系的行命并降低了 Silent Failure 的生率，但并不能消除底 Intelligent Entropy Principle。在极端任复度、超行周期（月）或极高Agent量的件下，序累最超越即使确定性治理也能承受的范。与 Intelligent Entropy Principle 的一致：外部束可以降低熵常 α 并提高故障值，但无法完全消除熵增。
--

Discussion

Theoretical Implications

理论意义

The Intelligent Entropy Principle reframes our understanding of LLM agent reliability:

Intelligent Entropy Principle 重新定义了我对LLM Agent可靠性的理解：

Gate Layer Theory: Entropy and Enforcement Persistence

Gate Layer Theory：熵与执行持久性

The Intelligent Entropy Principle has a direct corollary for how agent systems should be governed: any enforcement mechanism that depends on the agent's memory for its execution is itself subject to entropic decay. This observation yields a spectrum of three fundamentally distinct enforcement layers, distinguished by their dependence on agent state and the type of failure they prevent:

Intelligent Entropy Principle 对Agent系统的治理方式有一个直接推论：任何依赖Agent执行的执行机制本身都受到熵衰减的影响。这一洞察产生了三种根本不同的执行层，它们根据Agent的依赖程度和所防止的故障类型而区分：

The critical insight behind this three-layer taxonomy is that each gate type addresses a failure mode the others cannot catch. Memory gates decay with agent state; logic gates catch structural errors invisible to both agent awareness and physical verification; physical gates enforce deterministic constraints that neither awareness nor logical analysis can guarantee. A complete enforcement architecture requires all three layers.

一三分法的洞在于，每种型都其他法捕的故障模式。Memory Gates Agent衰；Logic Gates 捕Agent意和物理均不可的构性；Physical Gates 行意和 分析都法保的确定性束。一完整的行架构需要全部三。

This observation yields the Irreversible Protection Principle: any operation with irreversible physical side effects must be protected by a physical gate, never solely by a memory or logic gate. Furthermore, any decision with complex preconditions should be protected by a logic gate before reaching the physical gate layer. The companion work on Agent Delivery Engineering [liu2025ade] formalizes this as Gate Layer Theory and maps the full ADE protocol stack across all three enforcement layers.

一察生了 Irreversible Protection Principle（不可逆保原）：任何具有不可逆物理副作用的操作必由 Physical Gate 保，不能依 Memory Gate 或 Logic Gate。此外，任何具有复前置件的策在到 Physical Gate 之前由 Logic Gate 保。Agent Delivery Engineering 的姊妹工作 \cite{liu2025ade} 此形式化 Gate Layer Theory，并完整的ADE映射到全部三行。

Practical Implications for System Design

系意的

Several design implications follow directly:

以下几意直接得出：

Relationship to Prior Work

与先前工作的系

The study of failures in LLM-based systems has largely been driven by security concerns. Prompt injection attacks [perez2022, greshake2023] represent a significant body of work, demonstrating that adversarial inputs can subvert LLM behavior. However, these studies focus on externally triggered failures—attacks that require an active adversary. Our work addresses a distinct class: failures that occur without any external trigger, under normal operating conditions.

基于LLM系□中故障的研究主要由安全□切□□。Prompt注入攻□ \cite{perez2022, greshake2023} 代表了大量工作，□明了□抗性□入可以□覆LLM行□。然而，□些研究□注的是外部触□的故障——需要主□攻□者的攻□。我□的工作□□的是一□不同的□□：在正常□行□件下、□需任何外部触□而□生的故障。

Research on RAG (Retrieval-Augmented Generation) quality degradation [lewis2020, shuster2021] has documented retrieval failures, context window overflow, and document selection errors that can degrade output quality. While these share surface similarities with silent failures—disorder without explicit errors—the root cause is architectural rather than structural. RAG quality issues can be addressed through improved retrieval, better chunking, or expanded context windows. Silent failures, conversely, stem from intrinsic properties of language-based autonomous reasoning and cannot be eliminated through architectural optimization alone.

□于RAG (Retrieval-Augmented Generation, □索增□生成) □量退化的研究 \cite{lewis2020, shuster2021} □□了□索失□、上下文窗□溢出和文□□□□等可能降低□出□量的□□。□然□些□□与 Silent Failure 在表面上有相似之□——□□式□□的□序——但根本原因是架构性的而非□构性的。RAG□量□□可以通□改□□索、更好的分□或□展上下文窗□□解□。而 Silent Failure □源于基于□言的自主推理的□在□性，□法□通□架构优化□消除。

The broader field of software reliability engineering [musa2004, lyu2007] provides mature frameworks for modeling and predicting failures in deterministic systems. However, these frameworks assume that system behavior is governed by deterministic rules with well-defined failure modes. LLM agent systems operate probabilistically at every level—from token generation to memory retrieval to task planning—rendering classical reliability models inapplicable to the emergent failure patterns we document.

更□泛的□件可靠性工程□域 \cite{musa2004, lyu2007} □建模和□□确定性系□中的故障提供了成熟框架。然而，□些框架假□系□行□由具有明确定□故障模式的确定性□□支配。LLM Agent 系□在每□□面——□Token生成到□□□索再到任□□划——都以概率方式□行，使得□典可靠性模型不适用于我□□□的涌□故障模式。

Most relevant to our work is the concurrent discovery of Library Drift [zhang2026], which identifies unbounded skill accumulation in self-evolving agent libraries as a silent failure mode. While Library Drift focuses on skill lifecycle management in single-agent ratchet architectures, our work addresses the broader class of silent failures across multi-agent systems—including cross-agent communication, memory persistence, and systemic entropy—and provides a unified theoretical framework (the Intelligent Entropy Principle) that subsumes both phenomena.

与我□工作最相□的是 Library Drift 的同期□□ \cite{zhang2026}，□工作□自□化Agent技能□中的□界技能累□□□一种 Silent Failure 模式。□然 Library Drift 聚焦于□Agent棘□架构中的技能生命周期管理，我□的工作□□跨多Agent系□的更□泛 Silent Failure □□——包括跨Agent通信、□□持久化和系□熵——并提供了一□□一的理□框架（Intelligent Entropy Principle），□□种□象□入其中。

Our previous work on Channel Fracture [liu2025channel] established the empirical foundation for the current theoretical framework, demonstrating cross-agent communication decay through controlled experimentation. The present work extends these findings into a general law governing entropy in LLM agent systems, with the 22-premise derivation providing a theoretical explanation for the empirical patterns observed. The ADE protocol suite and PIG Engine [liu2025ade] provide the engineering countermeasure validated by the experiments reported here.

我□先前□于 Channel Fracture 的工作 \cite{liu2025channel} □□前理□框架奠定了□□基□，通□受控□□□明了跨Agent通信衰□。本工作□□些□□□展□一□支配LLM Agent系□中熵的普遍定律，22□前提的推□□察到的□□模式提供了理□解□。ADE□□套件和PIG Engine \cite{liu2025ade} 提供了□本□告□□□□的工程□策。

We note that the concept of "entropy" in LLM systems has been used informally in prior discussions of model drift and output variability [holtzman2020, zhang2026]. Our contribution is to formalize this concept into a measurable, predictive quantity $S(t)$ with an experimentally validated exponential growth law and a decomposable entropy constant α .

我注意到，"熵"这一概念在LLM系中已在模型漂移和输出异性的先前中被非正式使用 \cite{holtzman2020, zhang2026}。我的在于将此概念形式化——可量、可的量 $S(t)$ ，并附增定律和可分解的熵常 α 。

Conclusion

This paper has presented the Intelligent Entropy Principle of LLM Agent Systems: a formalization of the observation that language-based autonomous systems, operating without external deterministic constraints, experience monotonic entropy increase as a function of interaction rounds. We have identified 22 intrinsic properties spanning the agent lifecycle that jointly entail this law, measured the entropy constant α across multiple architectures, and presented the PIG+ADE engineering countermeasure as a production-validated approach to managing entropy-driven disorder.

本文提出了LLM Agent系的 Intelligent Entropy Principle：以下观察的形式化——基于语言的自主系在有外部确定性约束的条件下运行，其熵作交互次的函数增加。我跨越了Agent生命周期的22在性，它共同含一定律；在多种架构下量了熵常 α ；并提出了PIG+ADE工程策作生熵的管理熵序的方法。

The central message is that silent failures are not implementation defects waiting to be patched. They are the behavioral expression of a structural property of LLM agent systems. The path forward is not the elimination of this property—which is impossible without changing the fundamental nature of language-based reasoning—but its management through deterministic governance layers that operate alongside, not within, the probabilistic agent execution path.

核心信息是：Silent Failure 不是等待修复的缺陷。它是LLM Agent系结构性性的行表现。前的道路不是消除——在不改基于语言的推理的根本性的前提下是不可能的——而是通在概率Agent行路之旁（而非之）行的确定性治理管理它。

AI Assistance Usage Declaration

AI□助使用□明

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